Dear Quillstone Payments AI Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. I have six years of experience in analytics and data science, including more than four years as a data scientist, building models and workflows that support decisions in a subscription business. I am particularly interested in applying my experience in forecasting, churn modelling and experimentation to AI systems that help merchants and support teams.

At Thistlebox, I built a weekly demand forecasting model using LightGBM and operationalised its results through a scheduled Vertex AI Workbench notebook writing to BigQuery. The model improved WAPE from 23% to 14%, with an estimated 9% reduction in food waste. I also developed a weekly churn propensity model, scored in BigQuery, whose retention targeting was validated through a 10% holdout. These projects have given me practical experience with feature engineering, batch workflows and using cloud-based data tools to put model outputs into operational use.

I have also written a Python and Streamlit experiment-planning app and its Dockerfile, and since 2025 have added pytest unit tests for forecasting feature code, reaching approximately 60% coverage. My production software engineering practice is developing, and I am keen to deepen it while working closely with backend engineers, data scientists and product teams. My experience is in predictive models and batch workflows rather than chat assistants or RAG, but I would bring a strong foundation in Python, statistical modelling and careful evaluation as I grow into those areas.

Quillstone's focus on helping merchants and its emphasis on team ownership, pragmatism and learning from mistakes appeal to me. I would welcome the opportunity to discuss how my experience could contribute to the AI team's work on customer support systems.

Kind regards,
Tobias Mwangi-Hartley